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BACKGROUND AND LIMITATIONS OF EXISTING APPROACHES

Many physiologic conditions and performance states that involve intimate-region function evolve across daily life and over extended time periods. However, conventional assessment approaches are often limited to snapshots collected during brief appointments, structured tests, or self-reported impressions. These snapshots may fail to capture variability driven by stress, circadian rhythm, recovery, hydration, vascular dynamics, musculoskeletal tension, and other context factors that influence intimate-region signals in real-world conditions.

Existing consumer approaches frequently rely on subjective reporting, isolated single-session measurements, or outputs that are overly anatomical, explicit, or stigmatizing. Such approaches may be unsuitable for broad adoption, may discourage longitudinal tracking, and may not provide stable, privacy-preserving representations of change over time. In addition, many consumer devices are optimized for a single measurement modality and may not provide a coherent way to relate intimate-region signals to whole-body context, baseline structural parameters, or within-user history.

Clinical approaches may provide more objective measurements, but they are often constrained by limited frequency of collection, variability across environments, and the inability to capture extended real-world patterns. Further, many existing systems are not designed as modular multi-node architectures and therefore lack mechanisms for cross-device correlation between intimate-region signals and additional physiologic context signals. Without cross-device correlation, interpretation can be ambiguous, including difficulty distinguishing transient fluctuations from meaningful change, and difficulty attributing patterns to contextual drivers.

Additionally, existing systems generally do not support correlation between measurements captured from different users in a privacy-preserving manner. For example, conventional approaches do not provide a standardized framework for correlating outputs across devices used by different individuals, including correlation between a first user’s measured patterns and a second user’s measured patterns. This limitation applies across relationship contexts, including male-to-female correlation, male-to-male correlation, female-to-female correlation, and other pairings. Existing solutions also commonly lack neutral, non-explicit representations that permit comparison or correlation across individuals without requiring anatomical labeling or explicit imaging, and without requiring disclosure of sensitive details beyond abstracted metrics.

Accordingly, there is a need for systems and methods that enable multi-node intimate physiology sensing across time, support longitudinal pattern measurement under real-world conditions, provide privacy-preserving outputs using non-anatomical abstractions, and enable cross-device correlation to improve interpretation of patterns by referencing context signals, baseline acquisition, within-user history, and, in certain embodiments, correlation between users across a range of pairing contexts.